

WEEK 2 – POWER BI DATA ANALYSIS

Superstore Sales & Profit Analysis Dashboard

1. INTRODUCTION:

Microsoft Power BI is a powerful business intelligence tool used to analyze data and present information through interactive visualizations. This project focuses on analyzing the Superstore Sales Dataset to understand sales performance, profitability, customer orders, and regional business activities.

The dataset is processed and organized using Power Query and data modeling techniques. DAX (Data Analysis Expressions) is used to calculate key performance indicators such as Total Sales, Total Profit, Total Orders, Total Quantity, Average Sales, and Profit Margin.

Different visualizations, including KPI cards, column charts, line charts, slicers, and a decomposition tree, are used to develop an interactive dashboard. This project helps transform raw sales data into meaningful information that can be used for business analysis and decision-making.

2. OBJECTIVES:

The main objectives of this project are:

- To explore and analyze the Superstore Sales Dataset using Power BI.
- To clean, prepare, and organize the dataset for analysis.
- To create a structured data model using tables and relationships.
- To develop DAX measures for calculating business performance metrics.
- To examine sales and profit across product categories and regions.
- To visualize sales trends over time.
- To design an interactive dashboard with filtering options.
- To identify useful patterns and insights from the sales data.

3. DATASET DESCRIPTION:

The Superstore Sales Dataset is a business dataset that contains information about customer transactions, products, sales, profits, and geographical regions.

It is used to study different aspects of business performance, including product sales, customer segments, order quantities, and changes in sales over time.

Important Columns

Ship Date	Date when the order was shipped
Customer ID	Unique customer identification
Customer Name	Name of the customer

Product ID	Unique product identification
Product Name	Name of the product
Category	Main product category
Sub-Category	Specific product group
Sales	Revenue generated from sales
Quantity	Number of products ordered
Profit	Profit or loss from transactions
Region	Geographical sales region
Segment	Customer classification

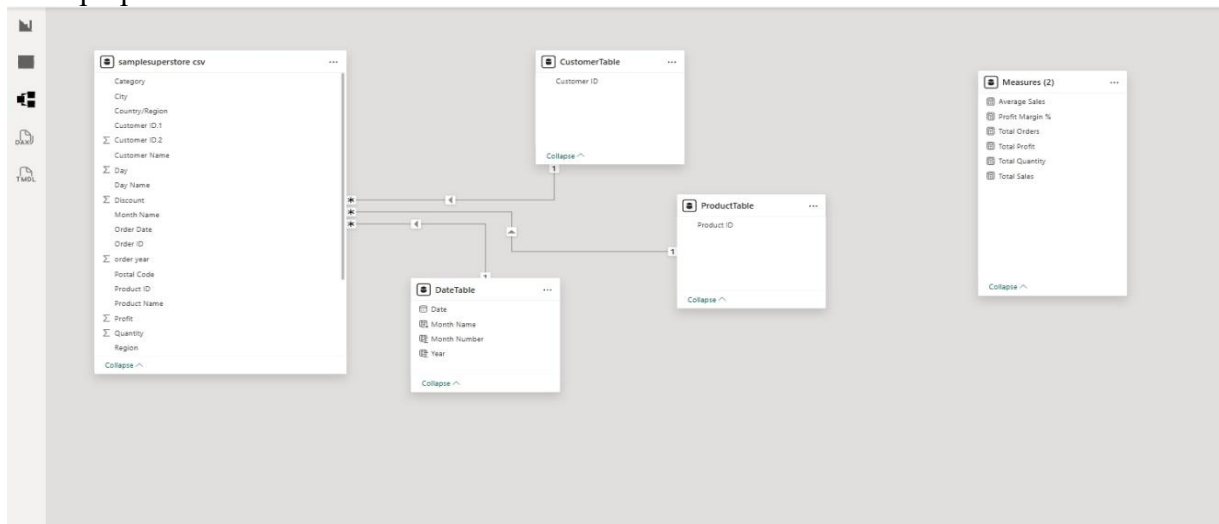
4. DATA PREPARATION:

The Superstore Sales Dataset was imported into Power BI for processing and analysis. The dataset was examined to understand its structure and identify the fields required for dashboard development.

Power Query was used to organize the data and verify the data types of important columns. Date fields were prepared for time-based analysis, while sales, profit, and quantity fields were used for numerical calculations.

A separate Date Table was created to support time-based analysis. Customer, Product, and Region tables were also organized as part of the data model. Relationships were established between the relevant tables to enable proper filtering and analysis.

The prepared dataset was then used to create DAX measures and build the final dashboard.



5. DATE TABLE CREATION:

A separate Date Table was created in Power BI to support time-based analysis and organize date information.

The table contains the following columns:

- Date
- Year
- Month Name
- Month Number

The Date Table was connected to the Order Date column in the Superstore dataset. This relationship enables the dashboard to display sales information across different years and months.

The Month Number column can also be used to arrange month names in chronological order, making the sales trend chart easier to understand

The screenshot shows a Power BI interface. On the left, a DAX measure is defined: `1 DateTable =`
`2 CALENDAR(`
`3 MIN('samplestore csv'[Order Date]),`
`4 MAX('samplestore csv'[Order Date]))`
`5 )`
Below the measure, a table is displayed with columns: Date, Year, Month Name, and Month Number. The table contains 26 rows of data for January 2023, starting from 01-01-2023 00:00:00 and ending at 26-01-2023 00:00:00. On the right, the 'Data' pane shows the hierarchy: Measures (2) > CustomerTable > DateTable > Date, with sub-items for Month Name, Month Number, Year, ProductTable, and samplestore csv.

Date	Year	Month Name	Month Number
01-01-2023 00:00:00	2023	January	1
04-01-2023 00:00:00	2023	January	1
05-01-2023 00:00:00	2023	January	1
06-01-2023 00:00:00	2023	January	1
07-01-2023 00:00:00	2023	January	1
08-01-2023 00:00:00	2023	January	1
09-01-2023 00:00:00	2023	January	1
10-01-2023 00:00:00	2023	January	1
11-01-2023 00:00:00	2023	January	1
12-01-2023 00:00:00	2023	January	1
13-01-2023 00:00:00	2023	January	1
14-01-2023 00:00:00	2023	January	1
15-01-2023 00:00:00	2023	January	1
16-01-2023 00:00:00	2023	January	1
17-01-2023 00:00:00	2023	January	1
18-01-2023 00:00:00	2023	January	1
19-01-2023 00:00:00	2023	January	1
20-01-2023 00:00:00	2023	January	1
21-01-2023 00:00:00	2023	January	1
22-01-2023 00:00:00	2023	January	1
23-01-2023 00:00:00	2023	January	1
24-01-2023 00:00:00	2023	January	1
25-01-2023 00:00:00	2023	January	1
26-01-2023 00:00:00	2023	January	1

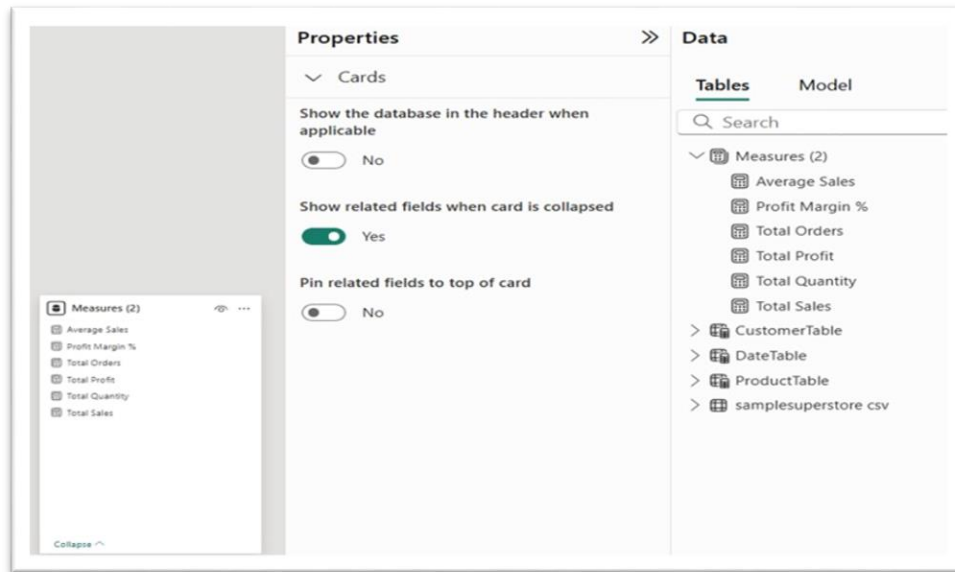
6. DAX MEASURES:

DAX stands for Data Analysis Expressions. It is used in Power BI to perform calculations and create measures that summarize business data.

The following measures were created to analyze the Superstore dataset.

Measure	Purpose
Total Sales	Calculates the overall sales amount
Total Profit	Calculates the total profit earned
Total Orders	Counts the number of orders
Total Quantity	Calculates the total quantity sold
Average Sales	Calculates the average sales value
Profit Margin %	Calculates profit as a percentage of sales

These measures are displayed using KPI cards and are also used in the dashboard visualizations to support interactive analysis.



7. DASHBOARD COMPONENTS:

The Superstore Sales and Profit Analysis Dashboard was developed using different Power BI visualizations. Each component provides a different view of the dataset and helps users explore business performance.

KPI Cards

Six KPI cards were designed to display important business metrics:

- Total Sales
- Total Profit
- Total Orders
- Total Quantity
- Average Sales
- Profit Margin %

These cards provide a quick summary of the selected data.

Category-wise Sales Analysis

A clustered column chart was created to compare sales across product categories.

This visualization helps users examine how sales are distributed among different categories and compare their contribution to overall sales.

Region-wise Profit Analysis

A clustered column chart was used to display profit across different geographical regions.

This chart allows users to compare regional profitability and identify variations in profit performance.

Sales Trend Analysis

A line chart was created to display sales over time using the Date Table.

It helps users observe changes in sales across different dates and identify possible trends or fluctuations.

Interactive Slicers

Slicers were added to make the dashboard interactive.

The dashboard includes filters for:

- Year
- Category
- Region
- Segment

Users can select different values to view specific portions of the dataset.

Decomposition Tree

A decomposition tree was included to explore the breakdown of Total Sales by Category.

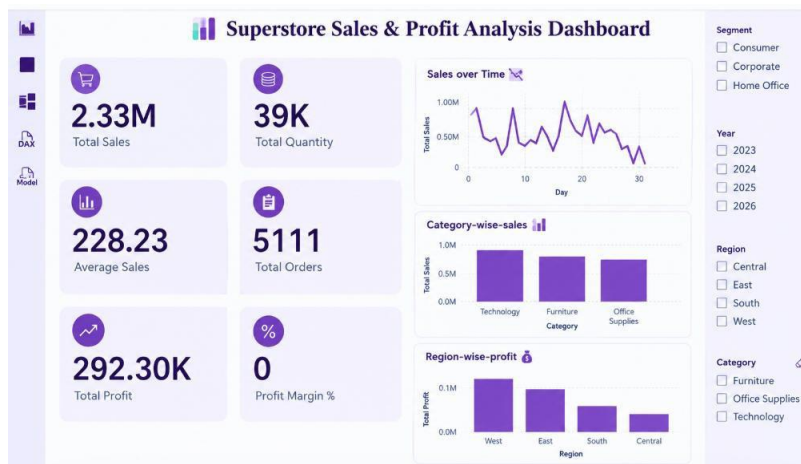
It allows users to expand the data into different categories and investigate how sales are distributed across the selected dimensions.

8. DASHBOARD DESIGN:

The final dashboard combines KPI cards, charts, slicers, and a decomposition tree into a single interactive report.

The layout is designed to present important business information in a clear and organized manner. Users can view overall performance through KPI cards and explore detailed information using charts and filters.

The dashboard provides different perspectives of sales and profit performance, making it easier to examine the dataset without manually reviewing individual records.



9. KEY INSIGHTS:

The dashboard provides the following types of business insights:

1. **Overall Sales Performance:** KPI cards summarize sales, profit, orders, and quantity for the selected data.
2. **Profitability Analysis:** Total Profit and Profit Margin % provide information about business profitability.
3. **Product Category Analysis:** Category-wise sales visualization helps compare the sales contribution of different product categories.
4. **Regional Performance:** Region-wise profit analysis highlights differences in profit across geographical regions.
5. **Sales Trends:** The line chart helps users examine sales movements over time.
6. **Customer Segment Analysis:** Segment slicers allow users to explore sales performance for different customer groups.
7. **Detailed Sales Breakdown:** The decomposition tree provides a way to investigate sales across different categories.

10. CONCLUSION:

The Superstore Sales and Profit Analysis project was developed using Microsoft Power BI to explore and visualize business sales data. The dataset was prepared and organized, and a structured data model was created using relevant tables and relationships.

DAX measures were used to calculate important business metrics, including Total Sales, Total Profit, Total Orders, Total Quantity, Average Sales, and Profit Margin. Various visualizations were combined to develop an interactive dashboard.

The completed dashboard enables users to examine sales performance, compare product categories, analyze regional profit, and explore sales trends over time. Overall, this project demonstrates the use of Power BI for data preparation, data modeling, business analysis, and interactive data visualization.

11. TOOLS AND TECHNOLOGIES:

Component	Technology Used
Business Intelligence Tool	Microsoft Power BI
Data Preparation	Power Query
Data Modeling	Tables and Relationships
Calculations	DAX
Data Visualization	KPI Cards, Column Charts, Line Chart
Interactive Analysis	Slicers and Decomposition Tree

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